

HANZHONG, China

WMO: 571270

Lat: 33.0677N

Lon: 107.0403E

Elev: 509

StdP: 95.36

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	-1.3	-9.7	1.7	5.0	-7.1	2.2	4.0	4.5	3.5	4.0	3.1	0.8	200	0.333

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	33.9	24.7	32.7	24.4	31.5	23.9	26.5	31.1	25.8	30.4	25.2	29.5	2.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	21.9	28.8	24.7	20.9	28.3	24.0	20.1	27.8	86.7	31.2	83.6	30.6	80.8	29.6	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.4	3.6	3.1	-4.4	35.9	1.1	1.4	-5.3	36.9	-5.9	37.7	-6.5	38.5	-7.4	39.5		
			-5.2	27.4	1.3	0.8	-6.1	28.0	-6.8	28.5	-7.5	29.0	-8.4	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	3.7	6.5	11.5	17.0	20.9	24.4	26.7	26.3	21.3	16.0	9.9	4.8
	DBStd	8.53	2.20	2.71	3.80	3.68	2.93	2.46	2.43	2.84	3.01	2.95	3.05	2.21
	HDD10.0	529	195	102	28	2	0	0	0	0	0	1	40	162
	HDD18.3	1842	453	332	213	68	11	0	0	0	7	85	254	421
	CDD10.0	2644	0	4	75	211	338	433	518	505	339	186	36	0
	CDD18.3	915	0	0	2	26	91	183	260	247	96	11	0	0
	CDH23.3	7590	0	0	14	187	598	1411	2529	2329	489	32	0	0
	CDH26.7	2746	0	0	1	36	158	481	1012	923	133	3	0	0
Wind	WSAvg	1.3	1.1	1.3	1.4	1.4	1.4	1.4	1.4	1.4	1.3	1.1	1.0	1.0
Precipitation	PrecAvg	899	8	11	34	70	96	98	178	125	166	76	32	8
	PrecMax	1463	32	39	86	219	278	372	381	452	360	181	83	33
	PrecMin	627	0	0	5	12	18	14	34	18	28	10	8	0
	PrecStd	193	7	10	19	41	57	69	83	93	98	45	19	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	17.4	25.1	29.8	32.1	33.7	35.7	35.6	32.7	26.3	19.1	13.0
		MCWB	6.8	9.9	15.4	18.9	20.5	22.4	25.6	25.3	23.9	18.5	13.9	8.0
	2%	DB	10.4	14.8	22.2	27.6	30.0	32.2	34.0	33.9	30.1	23.9	17.2	10.8
		MCWB	5.6	8.7	13.8	17.6	19.8	22.6	25.3	25.1	22.5	17.8	13.1	7.0
	5%	DB	8.7	12.7	20.0	25.5	28.4	31.0	32.7	32.5	27.9	22.0	15.5	9.5
		MCWB	4.9	7.7	12.8	16.9	19.4	22.5	25.0	24.8	21.5	17.3	12.3	6.4
	10%	DB	7.4	10.9	17.7	23.4	26.8	29.6	31.5	31.2	26.0	20.3	14.1	8.4
		MCWB	4.4	7.0	11.7	16.2	19.0	22.1	24.6	24.5	20.9	17.0	12.0	5.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	11.1	16.9	20.8	22.6	25.8	27.6	27.5	25.6	20.5	15.2	9.3
		MCDB	11.0	15.3	23.4	26.6	27.9	29.3	32.2	32.4	30.3	22.5	17.1	11.0
	2%	WB	6.3	9.5	14.7	19.3	21.6	24.7	26.7	26.5	24.0	19.1	14.1	8.2
		MCDB	9.2	13.2	20.0	24.8	26.8	28.8	31.2	31.3	27.1	21.8	15.8	9.7
	5%	WB	5.5	8.5	13.5	18.1	20.7	23.8	26.0	25.8	22.7	18.4	13.2	7.3
		MCDB	8.1	11.5	18.6	23.3	25.7	28.4	30.6	30.5	26.0	21.0	14.8	8.8
	10%	WB	4.7	7.6	12.4	17.0	20.0	23.1	25.4	25.2	21.7	17.5	12.3	6.3
		MCDB	7.0	10.2	16.9	22.3	25.0	27.6	29.8	29.6	24.8	19.8	13.8	7.9
Mean Daily Temperature Range	5% DB	MDBR	6.5	7.0	8.5	9.1	8.8	7.8	7.3	7.4	6.2	5.8	5.8	6.0
		MCDBR	9.4	11.1	12.9	13.1	12.2	10.9	9.5	9.5	9.7	8.7	8.5	8.1
		MCWBR	5.2	5.5	5.2	4.8	4.2	3.9	3.2	3.1	3.5	3.6	4.5	4.7
	5% WB	MCDBR	8.4	9.2	11.4	11.1	10.0	8.7	8.2	8.2	7.6	7.1	6.6	6.4
		MCWBR	5.0	4.9	5.0	4.9	4.2	3.7	3.2	3.1	3.1	3.2	3.9	4.0
Clear-Sky Solar Irradiance	taub	0.622	0.701	0.721	0.693	0.650	0.633	0.649	0.689	0.663	0.657	0.648	0.588	
	taud	1.540	1.441	1.446	1.531	1.663	1.741	1.738	1.639	1.662	1.604	1.565	1.618	
	Ebn at Noon	550	559	600	651	688	698	683	646	635	585	527	548	
	Edh at Noon	228	278	296	283	250	230	230	250	235	231	220	200	
All-Sky Solar Radiation	RadAvg	2.13	2.34	3.39	4.32	4.61	4.72	4.63	4.26	3.03	2.52	2.07	1.96	
	RadStd	0.30	0.46	0.34	0.54	0.50	0.43	0.74	0.58	0.41	0.48	0.36	0.33	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.40	+0.51	N/A	N/A	N/A	N/A	-55	-115	+106	+41
(47)	Regional (6 neighbors)	+0.27	N/A	N/A	N/A	N/A	N/A	-40	-90	+84	N/A

Nomenclature: See separate page